NL01CHEM01 Session 1 Problem Set Solutions

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§ 1 Solutions

Problem 1. **Element, Compound, or Mixture? (• ○ ○)**
Write E, C, or M next to each.

a) Copper wire (E)
Copper is one element.

b) Table salt (C)
Salt is the compound sodium chloride.

c) Well-stirred salt water (M)
Saltwater is a mixture.

d) Room air (M)
Air is mostly nitrogen and oxygen mixed.

e) Distilled water (C)
Distilled water is the compound H_2O.

Problem 2. **Pure, Uniform, or Non-Uniform? (• ○ ○)**
Label each as PURE, UNIFORM, or NON-UNIFORM.

a) Spoonful of stirred sugar water: UNIFORM
Stirred sugar water looks the same throughout but has sugar + water.

b) Handful of trail mix: NON-UNIFORM
Trail mix shows visible parts.

c) Sealed distilled water: PURE
Distilled water has fixed composition.

d) Rested Italian dressing: NON-UNIFORM
Rested dressing separates.

Problem 3. **A Mystery Block (• • ○)**
A small solid block has mass 54 g and volume 20 cm^3. (a) Compute its density. (b) Which of the following elements is the solid block most likely made of: aluminum (2.7 g/cm^3), iron (7.9 g/cm^3), or copper (8.9 g/cm^3)?

Notice that $\rho = 54/20 = 2.7\,\text{g/cm}^3$. Therefore, the block is most likely aluminum. Although color, shape, and size were not available, the measured density narrows down the choices.

Problem 4. Counting Parts of an Atom (• • ○)
An atom has atomic number 17 and mass number 35.

a) Which element is it?
chlorine ($Z = 17$)

b) Number of protons?
17 protons

c) Number of neutrons?
18 neutrons ($A - Z = 35 - 17$)

d) Number of electrons if neutral?
17 electrons for neutral atom

Problem 5. Average Atomic Mass (• • •)
An element has two isotopes. Isotope M weighs 10.0 u and makes up 20.0% of atoms. Isotope N weighs 11.0 u and makes up 80.0%. Find the average atomic mass to one decimal place.

The approximate average atomic mass can be represented as $(10.0)(0.20) + (11.0)(0.80) = 2.0 + 8.8 = 10.8$ u. This is because when multiplying each isotope mass by its fraction and adding the products, the total fractions sum must to 1.00.

Problem 6. Glass of Water (• • •)
Use 18 g/mol for water and 6.02×10^{23} particles/mol.

a) How many moles in 180 g?
$180/18 = 10$ mol

b) How many molecules total → write in scientific notation.
$10 \times 6.02 \times 10^{23} = 6.02 \times 10^{24}$ molecules

c) Roughly how many per sip if a glass is six sips?
About 1×10^{24} per sip